

Recursive Formulas Review

Directions: Show all work. No calculators.

1. Write a recursive formula to represent the sequence 8, 14, 20, 26, ...
2. Write a recursive formula to represent the sequence 25, 16, 7, -2 , ...
3. Write a recursive formula to represent the sequence -21 , -13 , -5 , 3, ...
4. Write a recursive formula to represent the sequence 54, 42, 30, 18, ...
5. Write the first 6 terms of a sequence if $a_1 = -9$ and $a_n = a_{n-1} + 8$.

6. Write the first 6 terms of a sequence if $a_1 = 44$ and $a_n = a_{n-1} - 9$.

7. Write the first 6 terms of a sequence if $a_1 = -28$ and $a_n = a_{n-1} + 6$.

8. Write the first 6 terms of a sequence if $a_1 = 13$ and $a_n = a_{n-1} - 7$.

9. A sequence has recursive formula $a_n = a_{n-1} - 5$ with $a_1 = 17$. What is the common difference for the sequence? Write the first 5 terms of the sequence.

10. A sequence has recursive formula $a_n = a_{n-1} + 10$ with $a_1 = -13$. What is the common difference for the sequence? Write the first 5 terms of the sequence.

11. A sequence has recursive formula $a_n = a_{n-1} - 11$ with $a_1 = 30$. What is the common difference for the sequence? Write the first 5 terms of the sequence.
12. A sequence has recursive formula $a_n = a_{n-1} + 9$ with $a_1 = -25$. What is the common difference for the sequence? Write the first 5 terms of the sequence.
13. Convert $a_n = a_{n-1} + 7$ with $a_1 = -11$ to an explicit formula.
14. Convert $a_n = a_{n-1} - 6$ with $a_1 = 22$ to an explicit formula.
15. Convert $a_n = a_{n-1} - 8$ with $a_1 = -2$ to an explicit formula.

16. Convert $a_n = a_{n-1} - 10$ with $a_1 = 38$ to an explicit formula.
17. An arithmetic sequence has $a_1 = 6$ and $a_7 = 36$. Find the common difference and write a recursive formula for the sequence.
18. An arithmetic sequence has $a_1 = -10$ and $a_5 = 18$. Find the common difference and write a recursive formula for the sequence.
19. An arithmetic sequence has $a_1 = 15$ and $a_9 = -17$. Find the common difference and write a recursive formula for the sequence.
20. An arithmetic sequence has $a_1 = -4$ and $a_6 = 31$. Find the common difference and write a recursive formula for the sequence.